M150W26 LECTURE 1.2: VARIETIES OVER FINITE FIELDS

Contents

Lecture 1.2: Varieties over Finite Fields

- Topics: Finite fields $\mathbb{F}_q$ and their extensions, $\mathbb{F}_q$-rational points of affine and projective varieties, definition of the zeta function of a variety over $\mathbb{F}_q$ or $\mathbb{Z}$, basic example of $\mathbb{A}^n$ and $\mathbb{P}^n$, functoriality and product formulas.
- References: [Mus11, § 2.1-2.3], [Poo17, § 2]

1. Finite Fields $\mathbb{F}_q$ and their extensions

Proposition 1.1. *Recall the following facts about Galois theory of finite fields:*

- *For every power of a prime q^r there is a unique field of order q^r up to isomorphism, the splitting field of $x^{q^r} - x$ over $\mathbb{F}_q$.*
- *The* (arithmetic) Frobenius *σ is a $\mathbb{F}_q$-automorphism of any algebraic extension of $\mathbb{F}_q$ given by $\sigma(x) = x^q$.*
- *The Galois group $\mathrm{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q)$ is isomorphic to $\mathbb{Z}/r\mathbb{Z}$ and is generated by σ.*
- *The field $\overline{\mathbb{F}}_q$ contains exactly one subfield of order q^r for every positive integer r, with elements given by*
$$\mathbb{F}_{q^r} = \{x \in \overline{\mathbb{F}}_q | \sigma^r(x) = x\}$$
- *The absolute Galois group is the following inverse limit: $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \cong \varprojlim \mathrm{Gal}(\overline{\mathbb{F}}_{q^r}/\mathbb{F}_q) \cong \varprojlim \mathbb{Z}/r\mathbb{Z} \cong \widehat{\mathbb{Z}}$*
- *σ is a topological generator for $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$.*

2. K-Rational Points on Varieties

Proposition 2.1. *Let L/K be a field extension, and let X be a scheme over K. Then*
$$X(L) \cong \bigsqcup_{x \in X} \mathrm{Hom}_{K-alg}(k(x), L)$$

Definition 2.2. Let a *variety over K* be a reduced, separated scheme of finite type over K.

Proposition 2.3. *If L/K is an algebraic extension and X is a variety, then in particular*
$$X(L) \cong \bigsqcup_{x \in X_{\mathrm{cl}}} \mathrm{Hom}_{K-alg}(k(x), L)$$

Proof. If x is not a closed point, then $\dim \overline{\{x\}} \geq 1$, but $\dim \overline{\{x\}} = \mathrm{trdeg}_K(k(x))$, so $k(x)$ is transcendental. Thus, we need not consider non-closed points in the disjoint union. $\square$

Lemma 2.4. *If X is a variety over K, then for all $x \in X_{\mathrm{cl}}$, $k(x)/K$ is a finite extension.*

Proof. The point x is contained in an affine patch isomorphic to $\mathrm{Spec}\, B$ with B a finitely generated K-algebra because X is finite type over K, and corresponds to a maximal ideal $\mathfrak{m} \subset B$. We know that $\mathscr{O}_{X,x} \cong B_{\mathfrak{m}}$, and
$$k(x) = \mathscr{O}_{X,x}/\mathfrak{m}_x \cong B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}.$$
We know, however, that $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} \cong \mathrm{Frac}(B/\mathfrak{m})$, but $B/\mathfrak{m}$ is a field, so $k(x) = B/\mathfrak{m}$. Finally, by Nullstellensatz, $k(x)$ is a finite field extension of K hence isomorphic to $K[x]/(f(x))$ for some irreducible polynomial f. $\square$

Proposition 2.5. *Let X be a variety over K and $x \in X_{\mathrm{cl}}$ with $k(x) \cong K[x]/(f(x))$. Then*

$$\mathrm{Hom}_{K-alg}(k(x), L) \leftrightarrow \{y \in L | f(y) = 0\}$$

Proof. A K-algebra homomorphism $K[x]/(f(x))$ is determined by where it maps x, and x must be sent to a root of $f(x)$ in L. □

Proposition 2.6. *Let X be an affine variety over K, i.e. $X = \operatorname{Spec} K[x_1, ..., x_n]/(f_1, ..., f_k)$ where $f_1, ..., f_k$ are polynomials in $x_1, ..., x_n$. Then an L-rational point of K equivalent to a tuple $(\alpha_1, ..., \alpha_n) \in L^n$ such that $f_i(\alpha_1, ..., \alpha_n) = 0$ for all i.*

Proof. An L-rational point of X is a K-scheme morphism $\operatorname{Spec} L \to \operatorname{Spec} K[x_1, ..., x_n]/(f_1, ..., f_k)$, which is equivalent to a K-algebra morphism $\phi : K[x_1, ..., x_n]/(f_1, ..., f_k) \to L$. This morphism is uniquely determined by any choice of $\phi(x_1), ..., \phi(x_n)$ that satisfies $f_i(\phi(x_1), ..., \phi(x_n)) = 0$ for all i, so take $\alpha_j = \phi(x_j)$ for all j. □

Proposition 2.7 (Basic Properties/Functoriality of $X(K)$). *Let L/K be a field extension.*

(a) *Taking L-points of a K-scheme defines a functor* $\mathrm{Sch}/K \to \mathrm{Sets}$.
(b) *If U_i is an open cover of X, then $\bigcup_i U_i(L) = X(L)$.*
(c) *For any schemes X, Y over K, $X \times Y(L) = X(L) \times Y(L)$ for any extension L of K.*

Proof.

(a) Given a morphism of K-schemes $f : X \to Y$, define a map of sets $f(L) : X(L) \to Y(L)$ by $g \mapsto f \circ g$.
(b) The topological image of a morphism $f : \operatorname{Spec} L \to X$ is a single point, and $f_* \mathscr{O}_{\operatorname{Spec} L}$ is determined by its stalk at that point, so the morphism is fully specified by a morphism to an affine open neighborhood of the image point.
(c) Compare universal properties.

□

3. $\mathbb{F}_q$-rational points of Varieties

First, we build a characterization of the $\mathbb{F}_q$-rational points of a variety using the closed points of the variety.

Let the degree of a closed point of X be $\deg(x) = [k(x) : K]$.

Proposition 3.1. *Let X be a variety over $\mathbb{F}_q$. Then*

$$|X(\mathbb{F}_{q^r})| = \sum_{e|r} e|\{x \in X_{\mathrm{cl}} | \deg(x) = e\}|$$

Proof. For each closed point $x \in X_{\mathrm{cl}}$, we need to determine $|\mathrm{Hom}_{\mathbb{F}_q-alg}(k(x), L)|$. Assume that $\deg(x) = e$. Then $k(x)$ is a finite field of order q^e, i.e. $\mathbb{F}_{q^e}$, so

$$|\mathrm{Hom}_{\mathbb{F}_q-alg}(k(x), L)| = |\mathrm{Hom}_{\mathbb{F}_q-alg}(\mathbb{F}_{q^e}, \mathbb{F}_{q^r})| = |\mathrm{Gal}(\mathbb{F}_{q^e}/\mathbb{F}_q)|.$$

This quantity is e if $e|r$ and 0 otherwise. Thus,

$$|X(\mathbb{F}_{q^r})| = \sum_{x \in X_{\mathrm{cl}} \text{s.t.} \deg(x)|r} \deg(x) = \sum_{e|r} e|\{x \in X_{\mathrm{cl}} | \deg(x) = e\}|.$$

□

Second, we adopt a more geometric viewpoint and view the $\mathbb{F}_q$-rational points as the intersection of subvarieties of $\overline{X} \times \overline{X}$ where $\overline{X} = X \times_{\operatorname{Spec} \mathbb{F}_q} \operatorname{Spec} \overline{\mathbb{F}}_q$. First, we define Frobenius morphisms on a scheme.

Definition 3.2. The *(absolute) Frobenius morphism* on a scheme X over $\mathbb{F}_q$ is denoted $\mathrm{Frob}_{X,q} : X \to X$ and given by the identity on the topological space X and by $f \mapsto f^q$ on $\mathscr{O}_X$.

Theorem 3.3. *Let X be a variety over $\mathbb{F}_q$. The natural embedding $X(\mathbb{F}_{q^r}) \hookrightarrow X(\overline{\mathbb{F}}_q)$ identifies $X(\mathbb{F}_{q^r})$ with the elements of $X(\overline{\mathbb{F}}_q)$ fixed by $\mathrm{Frob}_{X,q}^r$.*

Proof. First consider the case that $X = \operatorname{Spec} A$ for some finitely generated $\mathbb{F}_q$-algebra A. Let $f : \operatorname{Spec} \mathbb{F}_{q^r} \to \operatorname{Spec} A$ be an $\mathbb{F}_{q^r}$-point of X with closed point x. Under the natural embedding this is mapped to the $\overline{\mathbb{F}}_q$ point $f \circ g$ where $g : \operatorname{Spec} \overline{\mathbb{F}}_q \to \operatorname{Spec} \mathbb{F}_{q^r}$ is the natural map coming from the inclusion of fields.

The action of $\operatorname{Frob}^r_{X,q}$ on $X(\overline{\mathbb{F}}_q)$ is given by postcomposing a point with $\operatorname{Frob}^r_{X,q}$.

Equivalently, we could take a $\overline{\mathbb{F}}_q$-point to be a $\mathbb{F}_q$-algebra homomorphism $f' : A \to \overline{\mathbb{F}}_q$ and the action of $\operatorname{Frob}^r_{X,q}$ precomposes by the map $F : A \to A$ given by $a \mapsto a^{qr}$. Then the $\overline{\mathbb{F}}_q$-points which are in fact $\mathbb{F}_{q^r}$-points are those for which f' factors through the inclusion $\mathbb{F}_{q^r} \hookrightarrow \overline{\mathbb{F}}_q$. We know that $A \cong \mathbb{F}_q[x_1, ..., x_n]/(f_1, ..., f_k)$, so f' is just the choice of $f'(x_1), ..., f'(x_n) \in \overline{\mathbb{F}}_q$ satisfying $f_1, ..., f_k$, so factoring through the inclusion just means that $f'(x_1), ..., f'(x_n) \in \mathbb{F}_{q^r}$. Moreover, f' is fixed by $\operatorname{Frob}^r_{X,q}$ if and only if $f'(x_i^{qr}) = f'(x_i)$ for all $i = 1, ..., n$. All of this arithmetic is happening mod p for some $p|q$, however, so $f'(x_i^{qr}) = (f'(x_i))^{qr}$.

To handle the case where X is not affine, repeat the above argument on an affine open cover. $\square$

Corollary 3.4. *Let $\Delta, \Gamma_r \subset \overline{X} \times \overline{X}$ be the diagonal and the graph of $\operatorname{Frob}^r_{X,q} \times \mathrm{id}$, respectively, then $X(\mathbb{F}_q^r)$ is in natural bijection with the closed points of $\Gamma_r \cap \Delta$.*

4. Zeta Function of a variety over $\mathbb{F}_q$

Definition 4.1. Logarithm and Exponential Power Series:

$$\exp(t) = \sum_{m \geq 0} \frac{t^n}{n!} \qquad \log(1+t) = \sum_{m \geq 1} \frac{(-1)^{m+1} t^m}{m}$$

Definition 4.2. Let X be a variety over a finite field $\mathbb{F}_q$, and define $N_m = |X(F_{q^m})|$ for all $m \geq 1$. The *(Hasse-Weil) zeta function* of X is

$$Z(X,t) = \exp\left(\sum_{m \geq 1} \frac{N_m}{m} t^m \right) \in \mathbb{Q}[[t]]$$

Proposition 4.3. *For every variety X over $\mathbb{F}_q$, we have*

$$Z(X,t) = \prod_{x \in X_{\mathrm{cl}}} \left(1 - t^{\deg(x)}\right)^{-1}$$

Proof. By Proposition 3.1,

$$\begin{aligned}
\log Z(X,t) &= \sum_{m \geq 1} \frac{N_m}{m} t^m \\
&= \sum_{m \geq 1} \frac{t^m}{m} \sum_{e|m} e|\{x \in X_{\mathrm{cl}} | \deg(x) = e\} \\
&= \sum_{m \geq 1} \frac{t^m}{m} \sum_{x \in X_{\mathrm{cl}} \text{s.t.} \deg(x)|r} \deg(x) \\
&= \sum_{x \in X_{\mathrm{cl}}} \sum_{\deg(x)|m} \frac{\deg(x) t^m}{m} \\
&= \sum_{x \in X_{\mathrm{cl}}} \sum_{a \geq 1} \frac{\deg(x) t^{a \deg(x)}}{a \deg(x)} \\
&= \sum_{x \in X_{\mathrm{cl}}} \sum_{a \geq 1} \frac{t^{a \deg(x)}}{a} \\
&= -\sum_{x \in X_{\mathrm{cl}}} \log(1 - t^{\deg(x)}),
\end{aligned}$$

so

$$Z(X,t) = \prod_{x\in X_{\mathrm{cl}}} \left(1 - t^{\deg(x)}\right)^{-1}$$

□

Corollary 4.4. $Z(X,t) \in \mathbb{Z}[[t]]$

Proof. Each element of the product is a geometric series with integer coefficients, so $Z(X,t)$ has integer coefficients as well. □

If we explicitly consider the geometric series $1 + t^{\deg(x)} + t^{2\deg(x)} + \cdots$ for each $x \in X_{\mathrm{cl}}$, we see that the coefficient of t^n in $Z(X,t)$ is the number of possible sets of points $x_1, ..., x_j$ with coefficients $a_1, ..., a_j$ such that

$$n = a_1 \deg(x_1) + \cdots + a_n \deg(x_n).$$

Definition 4.5. The *group of 0-cycles* $Z_0(X)$ is the free abelian group generated by X_{cl}. The *degree* of a 0-cycle $a_1x_1 + \cdots + a_nx_n$ is $a_1 \deg(x_1) + \cdots + a_n \deg(x_n)$.

Thus, we have proven the following proposition.

Proposition 4.6. *The zeta function $Z(X,t)$ is a generating function for the number of 0-cycles of degree n on X.*

5. Examples and Properties of Zeta Function

Example 5.1. The zeta function of affine space is given by

$$Z(\mathbb{A}^n_{\mathbb{F}_q}, t) = \frac{1}{1 - q^n t}$$

Proof. There are $|\mathbb{F}_{q^m}|^n = q^{mn}$ $\mathbb{F}_{q^m}$-rational points of $\mathbb{A}^n$ for every m, so

$$Z(\mathbb{A}^n_{\mathbb{F}_q}, t) = \exp\left(\sum_{m\geq 1} \frac{q^{mn}}{m} t^m\right) = \exp\left(-\log\left(1 - q^n t\right)\right) = \frac{1}{1 - q^n t}$$

□

Proposition 5.2. *If X is a variety over $\mathbb{F}_q$, Y is a closed subvariety, and $U = X \setminus Y$, then $Z(X,t) = Z(Y,t)Z(U,t)$.*

Proof. Observe that the closed points of X is the disjoint union of the closed points of Y and U, then use Proposition 4.3. □

Example 5.3. The zeta function of projective space is given by

$$Z(\mathbb{P}^n_{\mathbb{F}_q}, t) = \frac{1}{(1-t)(1-qt)\cdots(1-q^n t)}$$

Proof. Use the fact that $\mathbb{P}^n = \mathbb{P}^{n-1} \sqcup \mathbb{A}^n$ with $\mathbb{P}^{n-1}$ closed and $\mathbb{A}^n$ open. □

References

[Mus11] Mircea Mustaţă. Zeta functions in algebraic geometry, 2011. Available at https://public.websites.umich.edu/~mmustata/zeta_book.pdf. (document)

[Poo17] Bjorn Poonen. *Rational points on varieties*, volume 186 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2017. (document)